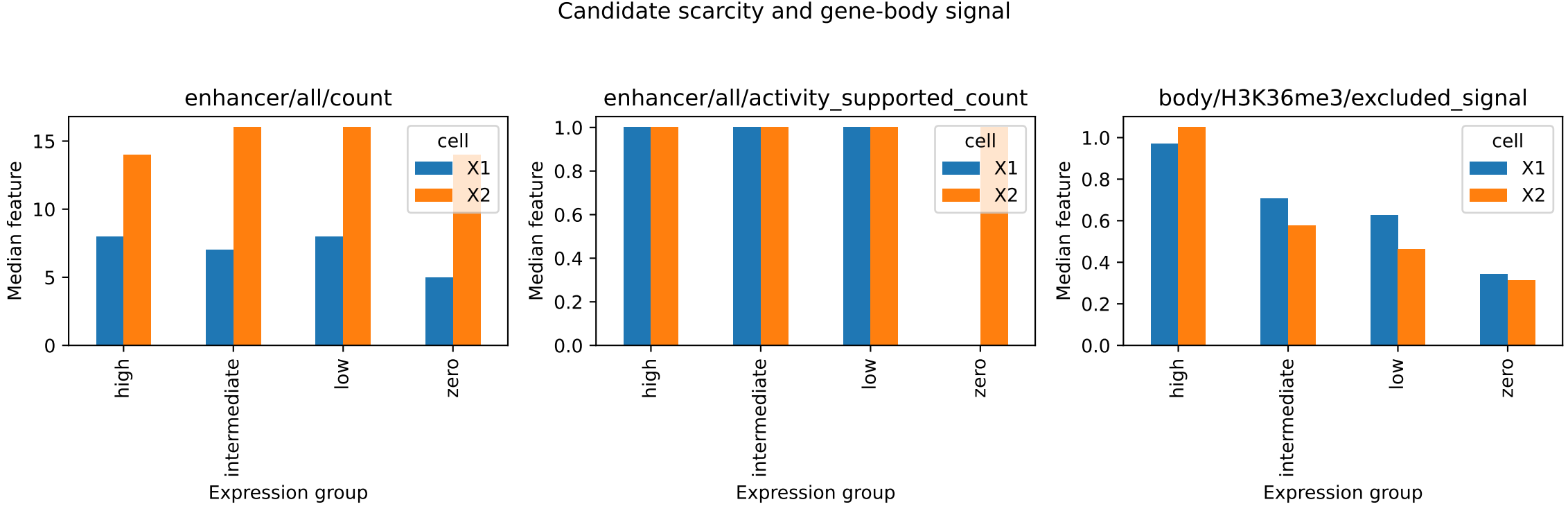
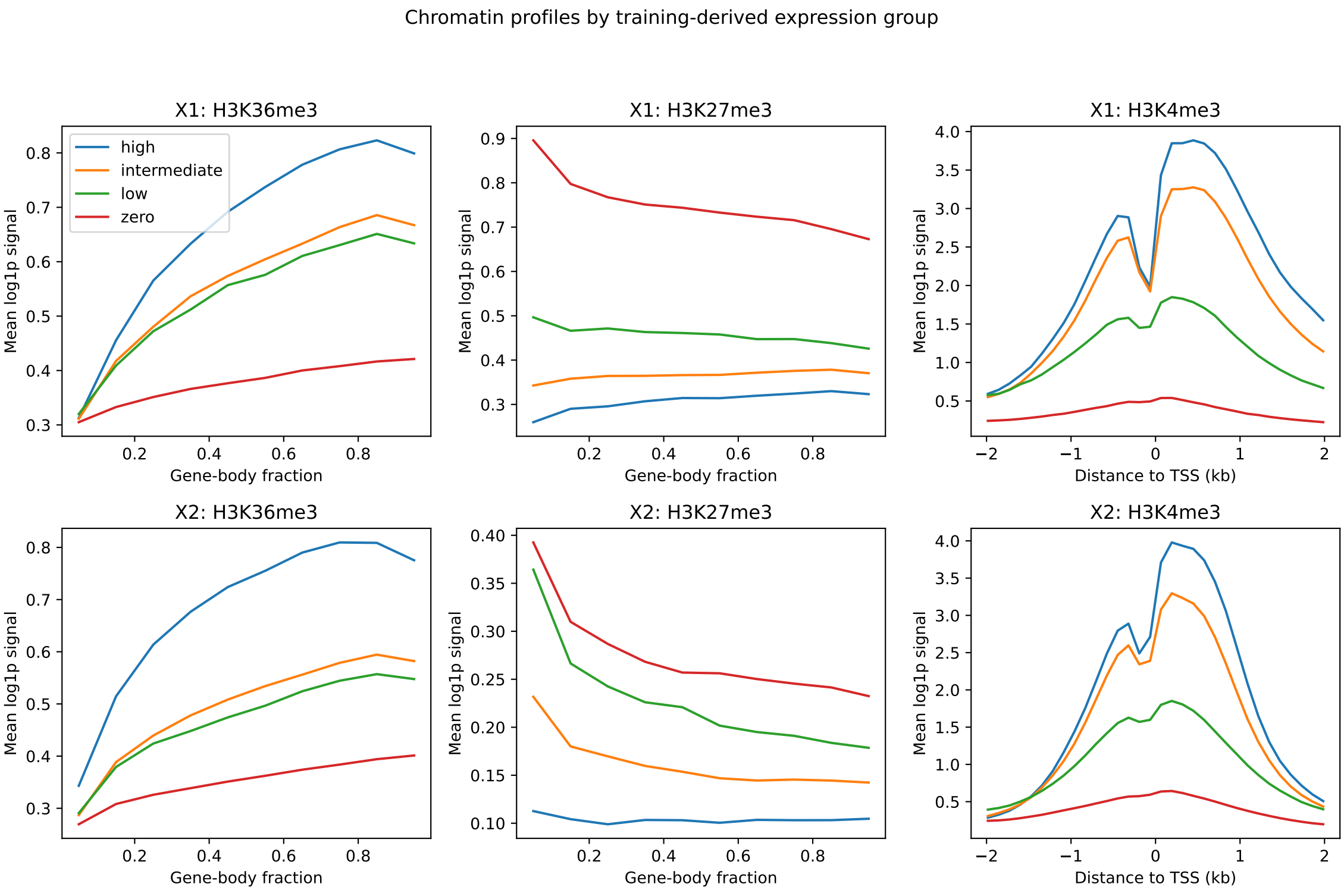
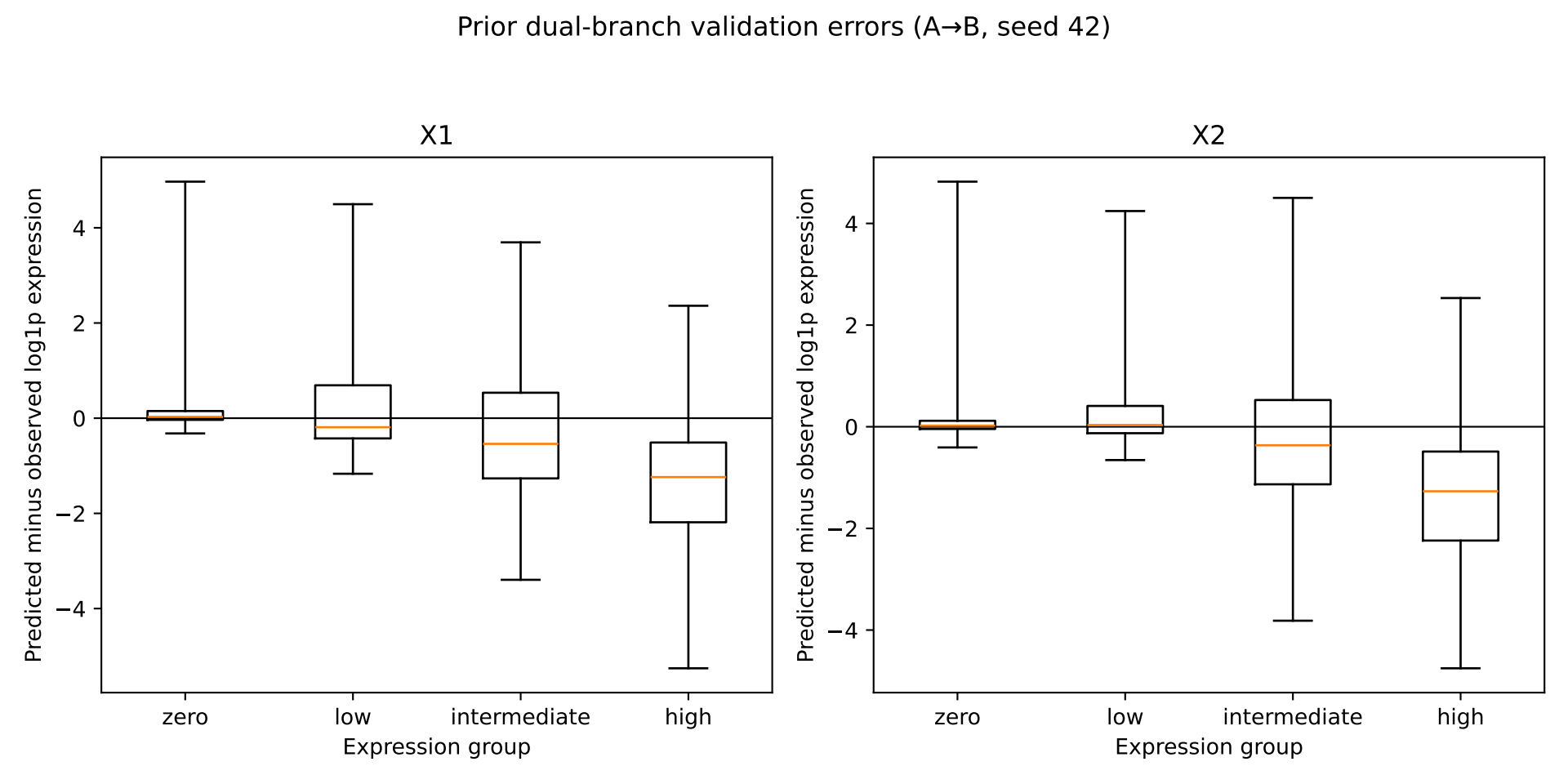
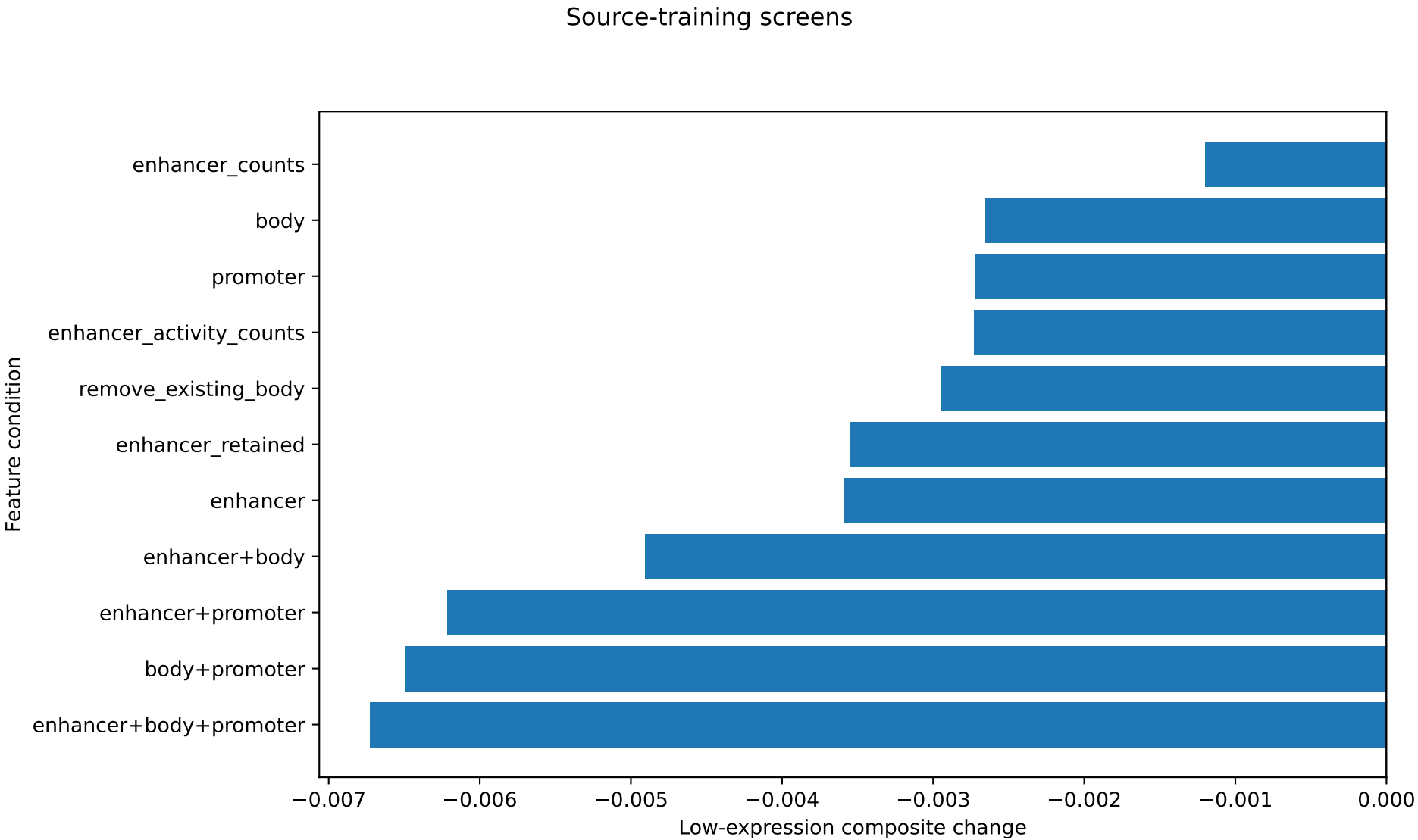
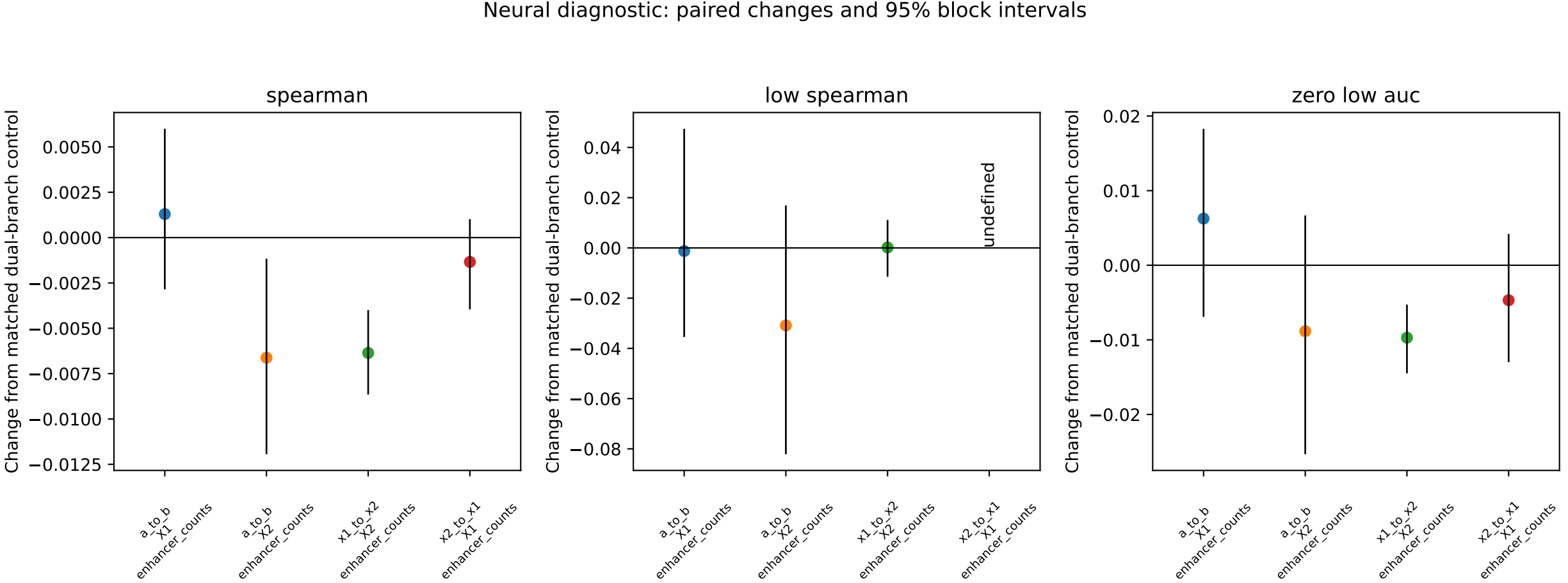
Expression groups (training-derived thresholds)
X1
X2
Genes
Expression group
high
intermediate
low
zero
0
2000
4000
6000
8000


Candidate scarcity and gene-body signal
enhancer/all/count
enhancer/all/activity_supported_count
body/H3K36me3/excluded_signal
Median feature
Expression group
high
intermediate
low
zero
cell
X1
X2
0
5
10
15
0.0
0.2
0.4
0.6
0.8
1.0


Chromatin profiles by training-derived expression group
X1: H3K36me3
X1: H3K27me3
X1: H3K4me3
X2: H3K36me3
X2: H3K27me3
X2: H3K4me3
high
intermediate
low
zero
Mean log1p signal
Gene-body fraction
Distance to TSS (kb)


Prior dual-branch validation errors (A→B, seed 42)
X1
X2
Predicted minus observed log1p expression
4
2
0
−2
−4
zero
low
intermediate
high
Expression group


Source-training screens
enhancer_counts
body
promoter
enhancer_activity_counts
remove_existing_body
enhancer_retained
enhancer
enhancer+body
enhancer+promoter
body+promoter
enhancer+body+promoter
Feature condition
−0.007
−0.006
−0.005
−0.004
−0.003
−0.002
−0.001
0.000
Low-expression composite change


Neural diagnostic: paired changes and 95% block intervals
spearman
low spearman
zero low auc
Change from matched dual-branch control
0.0050
0.0025
0.0000
−0.0025
−0.0050
−0.0075
−0.0100
−0.0125
0.04
0.02
0.00
−0.02
−0.04
−0.06
−0.08
undefined
0.02
0.01
0.00
−0.01
−0.02
a_to_b X1 enhancer_counts
a_to_b X2 enhancer_counts
x1_to_x2 X2 enhancer_counts
x2_to_x1 X1 enhancer_counts